

HW6 homework debrief.

Supp II

(i) A non-zero vector in $\mathbb{R}^n$ can sometimes be a part of an orthonormal basis for $\mathbb{R}^n$.

For instance, $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ is a part of an orthonormal basis for $\mathbb{R}^3$: $\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$.

But $\begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}$ is not part of any orthonormal basis for $\mathbb{R}^3$, since it is not normalized (It has length 2).

(ii) The product of two orthogonal matrices is always orthogonal.

If A and B are orthogonal, we have

$$A^{-1} = A^T \text{ and } B^{-1} = B^T.$$

Then AB is invertible (with inverse $B^{-1}A^{-1}$)

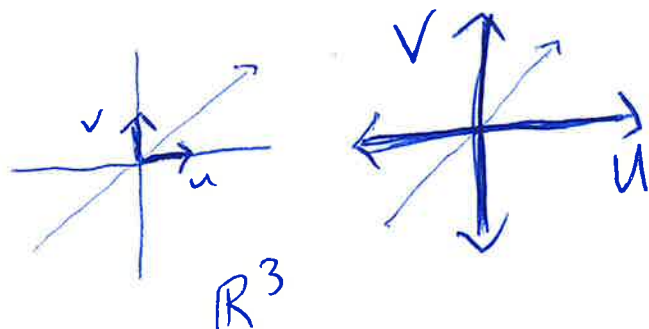
and has inverse $B^{-1}A^{-1} = B^T A^T = (AB)^T$

And since $(AB)^{-1} = (AB)^T$, we have that

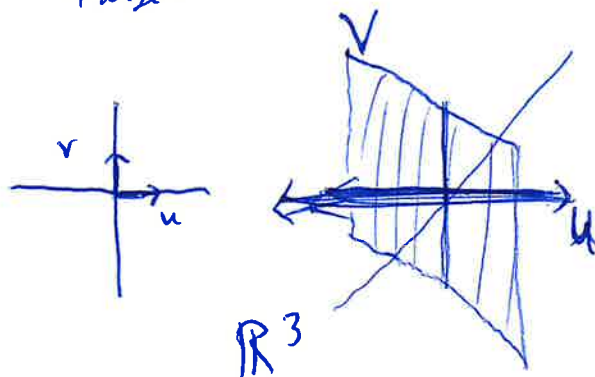
AB is orthogonal.

(iii) If two vectors are orthogonal, then a given pair of subspaces U and V containing u and v respectively are sometimes also orthogonal.

True:



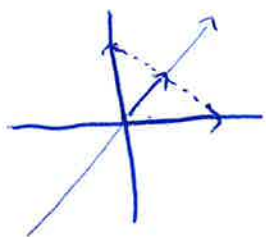
False:



5.3.9 (Using eigenvalues to compute determinants)

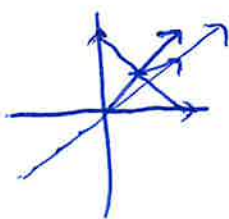
(a) Try small cases first:

$\mathbb{R}^2$



$$\begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}$$

$\mathbb{R}^3$



$$\begin{pmatrix} 1/3 & 1/3 & 1/3 \\ 1/3 & 1/3 & 1/3 \\ 1/3 & 1/3 & 1/3 \end{pmatrix}$$

$\mathbb{R}^n$

$$\begin{pmatrix} 1/n & \dots & 1/n \\ \vdots & \ddots & \vdots \\ 1/n & \dots & 1/n \end{pmatrix}$$

in each case, $\text{proj}_V(u) = \frac{u \cdot v}{v \cdot v} v$

so eg. $\frac{\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}}{\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1/3 \\ 1/3 \\ 1/3 \end{pmatrix}$

b)

$$\begin{pmatrix} 1 & \dots & \\ & \ddots & \\ & & 1/n \end{pmatrix} = n \begin{pmatrix} 1/n & \dots & \\ & \ddots & \\ & & 1/n \end{pmatrix}$$

and ~~values~~ ^{e-values of} $\begin{pmatrix} 1/n & \dots & \\ & \ddots & \\ & & 1/n \end{pmatrix} = 0 \text{ and } 1$

$\uparrow$ mult $\uparrow$ mult
 $n-1$ 1

so e-values of $\begin{pmatrix} 1 & \dots & \\ & \ddots & \\ & & 1/n \end{pmatrix}$ are 0 and n

$\uparrow$ $\uparrow$
 mult $n-1$ mult 1

c) e-values of $A - I$

$$\begin{pmatrix} 0 & & 1 \\ & \ddots & \\ 1 & & 0 \end{pmatrix}$$

Well, for every eigenvector of A , A stretches it by its eigenvalue, and then you subtract off one copy of that vector.

For v an eigenvector of A with eigenvalue λ :

$$(A - I)v = Av - Iv = \lambda v - v = (\lambda - 1)v$$

So since A has e-values 0 and n

$\uparrow$ $\uparrow$
 mult $n-1$ mult 1

$A - I$ has e-values -1 and $n-1$

$\uparrow$ $\uparrow$
 mult $n-1$ mult 1

d) $\det(A-I) =$ the product of its eigenvalues,
which is $(-1)^{n-1} \cdot (n-1)$